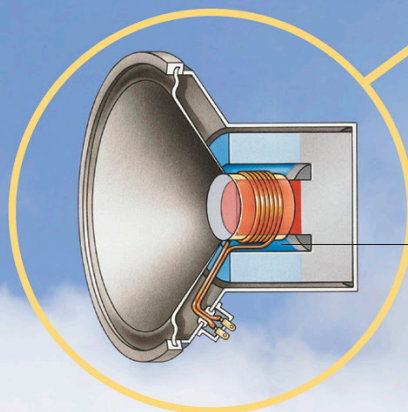


## Electromagnets

When an electric current flows through a wire coil, the coil becomes magnetic. This creates an electromagnet. Automatic doors, loudspeakers, and electric motors all use electromagnets.



Electromagnets are used in speakers.

## Lifting with magnets

Some cranes use magnetic force, in the form of giant electromagnets, instead of hooks. The electromagnet can be switched on or off.

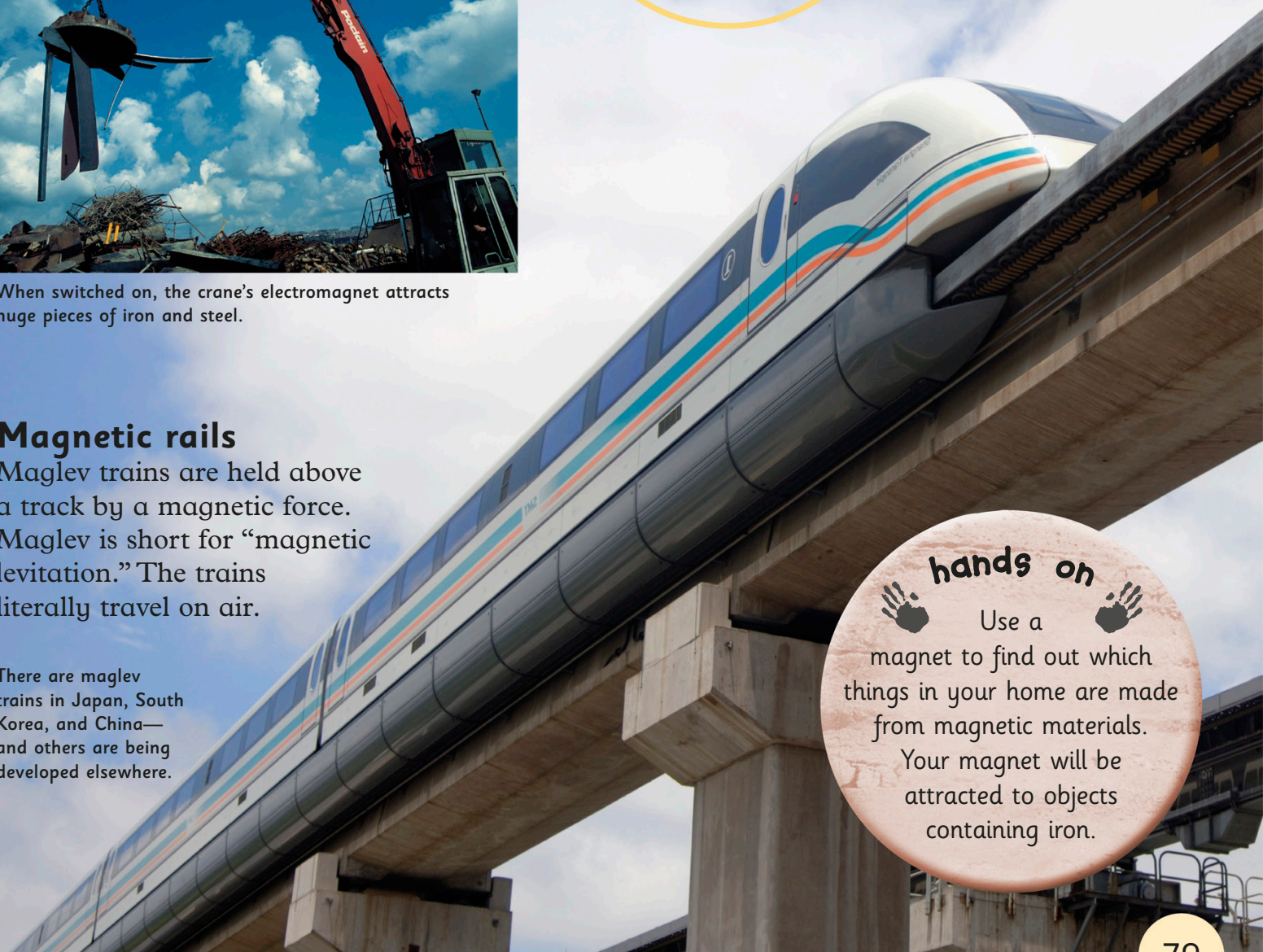


When switched on, the crane's electromagnet attracts huge pieces of iron and steel.

## Magnetic rails

Maglev trains are held above a track by a magnetic force. Maglev is short for “magnetic levitation.” The trains literally travel on air.

There are maglev trains in Japan, South Korea, and China—and others are being developed elsewhere.



**hands on**



Use a



magnet to find out which things in your home are made from magnetic materials.

Your magnet will be attracted to objects containing iron.